

Voyage Fuel Decision Report

Report year: 2026 | Departure: CNSHG | Arrival: NLRTM | Case currency: EUR

Boundary: voyage-level regulatory estimate for this submitted voyage. FuelEU indicative penalty equivalent is not a formal annual penalty or annual-limit settlement; execution conditions remain pending and this report is not a procurement recommendation.

Case boundary and scope

Field	Value
Departure EU ETS identity	THIRD_COUNTRY
Arrival EU ETS identity	IN_SCOPE
Departure FuelEU identity	THIRD_COUNTRY
Arrival FuelEU identity	IN_SCOPE
EU ETS reason	ONE_IN_SCOPE
EU ETS geographic scope rate	50.00
EU ETS surrender rate	100.00
FuelEU reason	ONE_IN_SCOPE
FuelEU scope rate	50.00
Source IDs	FACTOR-CATALOG:MDO; FACTOR-CATALOG:UCO_FAME; FACTOR-CATALOG:LNG_OTTO_MEDIUM_SPEED; PORT-02; ETS-01; FEU-01

Conclusions and conditional recommendations

ID	Condition	Status	Scenario	Reason / assumptions
CURRENT_MODEL_COST_MIN	CURRENT_MODEL_COST	CONDITIONAL	Ing-quote-1@0.1	MODEL_COST_MINIMUM; EXECUTION_CONDITIONS_PENDING; FEASIBLE; COMPARABLE
TARGET_MIN_COST:uco-quote-1	FUELEU_TARGET_MINIMUM_COST	CONDITIONAL	uco-quote-1@0.022130676682982508109158969772727924816074006340564	CONSTRAINT_RESULT; EXECUTION_CONDITIONS_PENDING; COMPARABLE
MAX_COMPLIANCE_IMPROVEMENT:uco-quote-1	FUELEU_MAX_COMPLIANCE_IMPROVEMENT	CONDITIONAL	uco-quote-1@0.3	CONSTRAINT_RESULT; EXECUTION_CONDITIONS_PENDING; COMPARABLE; TARGET_REACHABLE; FEASIBLE
REFERENCE_ADJUSTED_COST_SWITCH:uco-quote-1:1	REFERENCE_ADJUSTED_COST_SENSITIVITY	CONDITIONAL	-	LOWER_ENVELOPE_SWITCH; EXECUTION_CONDITIONS_PENDING; REFERENCE_VALUE_SENSITIVITY
TARGET_MIN_COST:Ing-quote-1	FUELEU_TARGET_MINIMUM_COST	UNAVAILABLE	-	TARGET_UNREACHABLE_UNDER_CONSTRAINTS; TARGET_UNREACHABLE_UNDER_CONSTRAINTS; EXECUTION_CONDITIONS_PENDING; COMPARABLE
MAX_COMPLIANCE_IMPROVEMENT:Ing-quote-1	FUELEU_MAX_COMPLIANCE_IMPROVEMENT	CONDITIONAL	Ing-quote-1@0.5	CONSTRAINT_RESULT; EXECUTION_CONDITIONS_PENDING; COMPARABLE; TARGET_UNREACHABLE_UNDER_CONSTRAINTS; FEASIBLE
REFERENCE_ADJUSTED_COST_SWITCH:Ing-quote-1:1	REFERENCE_ADJUSTED_COST_SENSITIVITY	CONDITIONAL	-	LOWER_ENVELOPE_SWITCH; EXECUTION_CONDITIONS_PENDING; REFERENCE_VALUE_SENSITIVITY

ID	Condition	Status	Scenario	Reason / assumptions
TARGET_MIN_COST:rfnbo-no-proof	FUELEU_TARGET_MINIMUM_COST	UNAVAILABLE	-	TARGET_NO_SOLUTION; TARGET_NO_SOLUTION; EXECUTION_CONDITIONS_PENDING; COMPARABLE
MAX_COMPLIANCE_IMPROVEMENT:rfnbo-no-proof	FUELEU_MAX_COMPLIANCE_IMPROVEMENT	UNAVAILABLE	-	SCENARIO_UNAVAILABLE; EXECUTION_CONDITIONS_PENDING; COMPARABLE; TARGET_NO_SOLUTION

Scenario comparison (fuel mass and energy)

Scenario	Candidate	Ratio	Baseline mass (t)	Candidate mass (t)	Energy (GJ)	Fuel cost	Model cost	CO2 (t)	CH4 (t)	N2O (t)	EUAs	EUA cost	Calc / constraint / execution
B0	B0	0.0000	100.000	0.000	4270.000	70000.00	83020.40	320.600000	0.005000	0.018000	162.755000	13020.40	CALCULABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
uco-quote-1@0.2	uco-quote-1	20.0000	82.194	20.549	4270.000	78084.70	91156.49	321.750048	0.005137	0.018494	163.397365	13071.79	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
uco-quote-1@0.3	uco-quote-1	30.0000	72.920	31.252	4270.000	82295.68	95394.24	322.349061	0.005209	0.018751	163.731947	13098.56	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
uco-quote-1@0.022 1306766829825081 0915896977272792 4816074006340564	uco-quote-1	2.2131	98.077	2.220	4270.000	70873.30	83899.25	320.724226	0.005015	0.018053	162.824387	13025.95	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
lng-quote-1@0.1	lng-quote-1	10.0000	88.671	9.852	4270.000	70444.16	83392.84	310.533139	0.309856	0.017011	161.858500	12948.68	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
lng-quote-1@0.5	lng-quote-1	50.0000	46.514	46.514	4270.000	72096.95	84778.75	273.073012	1.444265	0.013330	158.522502	12681.80	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
rfnbo-no-proof@0.1	rfnbo-no-proof	10.0000	90.000	10.000	4270.000	72000.00	85020.40	320.600000	0.005000	0.018000	162.755000	13020.40	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING
rfnbo-no-proof@0.2	rfnbo-no-proof	20.0000	80.000	20.000	4270.000	74000.00	87020.40	320.600000	0.005000	0.018000	162.755000	13020.40	COMPARABLE / FEASIBLE / EXECUTION_CONDITIONS_PENDING

Absolute and relative-to-B0 changes

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
B0	Baseline fuel mass	100.000	0.000	0.000000%
B0	Candidate fuel mass	0.000	0.000	-
B0	Physical energy	4270.000	0.000	0.000000%
B0	Fuel cost	70000.00	0.00	0.000000%
B0	CO2	320.600000	0.000000	0.000000%
B0	CH4	0.005000	0.000000	0.000000%
B0	N2O	0.018000	0.000000	0.000000%
B0	MRV raw CO2e	325.510000	0.000000	0.000000%
B0	EU ETS CO2e before scope	325.510000	0.000000	0.000000%
B0	EUAs required	162.755000	0.000000	0.000000%
B0	EUA cost	13020.40	0.00	0.000000%

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
B0	Model cost	83020.40	0.00	0.000000%
B0	FuelEU scoped energy	2135.000	0.000	0.000000%
B0	FuelEU RWD denominator	4270.000	0.000	0.000000%
B0	WtT	14.4000	0.0000	0.000000%
B0	TtW	76.3674	0.0000	0.000000%
B0	GHGI	90.7674	0.0000	0.000000%
B0	Target	89.3368	0.0000	0.000000%
B0	Compliance balance (g)	-3054432.000000	0.000000	0.000000%
B0	Compliance balance	-3.054432	0.000000	0.000000%
B0	Indicative penalty equivalent	1969.83	0.00	0.000000%
B0	Compliance improvement	0.000000	0.000000	-
B0	Reference-adjusted cost	-	-	-
uco-quote-1@0.2	Baseline fuel mass	82.194	-17.806	-17.805582%
uco-quote-1@0.2	Candidate fuel mass	20.549	20.549	-
uco-quote-1@0.2	Physical energy	4270.000	0.000	0.000000%
uco-quote-1@0.2	Fuel cost	78084.70	8084.70	11.549567%
uco-quote-1@0.2	CO2	321.750048	1.150048	0.358717%
uco-quote-1@0.2	CH4	0.005137	0.000137	2.743022%
uco-quote-1@0.2	N2O	0.018494	0.000494	2.743022%
uco-quote-1@0.2	MRV raw CO2e	326.794731	1.284731	0.394682%
uco-quote-1@0.2	EU ETS CO2e before scope	326.794731	1.284731	0.394682%
uco-quote-1@0.2	EUAs required	163.397365	0.642365	0.394682%
uco-quote-1@0.2	EUA cost	13071.79	51.39	0.394682%
uco-quote-1@0.2	Model cost	91156.49	8136.09	9.800105%
uco-quote-1@0.2	FuelEU scoped energy	2135.000	0.000	0.000000%
uco-quote-1@0.2	FuelEU RWD denominator	4270.000	0.000	0.000000%
uco-quote-1@0.2	WtT	0.8509	-13.5491	-94.090873%
uco-quote-1@0.2	TtW	76.6720	0.3046	0.398852%
uco-quote-1@0.2	GHGI	77.5230	-13.2445	-14.591677%
uco-quote-1@0.2	Target	89.3368	0.0000	0.000000%
uco-quote-1@0.2	Compliance balance (g)	25222559.578441	28276991.578441	925.769229%
uco-quote-1@0.2	Compliance balance	25.222560	28.276992	925.769229%
uco-quote-1@0.2	Indicative penalty equivalent	-	-	-
uco-quote-1@0.2	Compliance improvement	28.276992	28.276992	-
uco-quote-1@0.2	Reference-adjusted cost	-	-	-
uco-quote-1@0.3	Baseline fuel mass	72.920	-27.080	-27.079776%
uco-quote-1@0.3	Candidate fuel mass	31.252	31.252	-
uco-quote-1@0.3	Physical energy	4270.000	0.000	0.000000%
uco-quote-1@0.3	Fuel cost	82295.68	12295.68	17.565260%
uco-quote-1@0.3	CO2	322.349061	1.749061	0.545559%
uco-quote-1@0.3	CH4	0.005209	0.000209	4.171749%
uco-quote-1@0.3	N2O	0.018751	0.000751	4.171749%
uco-quote-1@0.3	MRV raw CO2e	327.463894	1.953894	0.600256%
uco-quote-1@0.3	EU ETS CO2e before scope	327.463894	1.953894	0.600256%

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
uco-quote-1@0.3	EUAs required	163.731947	0.976947	0.600256%
uco-quote-1@0.3	EUA cost	13098.56	78.16	0.600256%
uco-quote-1@0.3	Model cost	95394.24	12373.84	14.904575%
uco-quote-1@0.3	FuelEU scoped energy	2135.000	0.000	0.000000%
uco-quote-1@0.3	FuelEU RWD denominator	4270.000	0.000	0.000000%
uco-quote-1@0.3	WtT	-6.2062	-20.6062	-143.098927%
uco-quote-1@0.3	TtW	76.8307	0.4632	0.606598%
uco-quote-1@0.3	GHGI	70.6244	-20.1430	-22.191879%
uco-quote-1@0.3	Target	89.3368	0.0000	0.000000%
uco-quote-1@0.3	Compliance balance (g)	39950877.953647	43005309.953647	1407.964229%
uco-quote-1@0.3	Compliance balance	39.950878	43.005310	1407.964229%
uco-quote-1@0.3	Indicative penalty equivalent	-	-	-
uco-quote-1@0.3	Compliance improvement	43.005310	43.005310	-
uco-quote-1@0.3	Reference-adjusted cost	-	-	-
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Baseline fuel mass	98.077	-1.923	-1.923328%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Candidate fuel mass	2.220	2.220	-
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Physical energy	4270.000	0.000	0.000000%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Fuel cost	70873.30	873.30	1.247564%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	CO2	320.724226	0.124226	0.038748%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	CH4	0.005015	0.000015	0.296297%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	N2O	0.018053	0.000053	0.296297%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	MRV raw CO2e	325.648774	0.138774	0.042633%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	EU ETS CO2e before scope	325.648774	0.138774	0.042633%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	EUAs required	162.824387	0.069387	0.042633%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	EUA cost	13025.95	5.55	0.042633%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Model cost	83899.25	878.85	1.058590%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	FuelEU scoped energy	2135.000	0.000	0.000000%

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	FuelEU RWD denominator	4270.000	-0.000	-0.000000%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	WtT	12.9365	-1.4635	-10.163534%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	TtW	76.4003	0.0329	0.043083%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	GHGI	89.3368	-1.4306	-1.576168%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Target	89.3368	0.0000	0.000000%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Compliance balance (g)	0.000000	3054432.000000	100.000000%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Compliance balance	0.000000	3.054432	100.000000%
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Indicative penalty equivalent	-	-	-
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Compliance improvement	3.054432	3.054432	-
uco-quote-1@0.022130676682982508 1091589697727279248160740063405 64	Reference-adjusted cost	-	-	-
Ing-quote-1@0.1	Baseline fuel mass	88.671	-11.329	-11.329026%
Ing-quote-1@0.1	Candidate fuel mass	9.852	9.852	-
Ing-quote-1@0.1	Physical energy	4270.000	0.000	0.000000%
Ing-quote-1@0.1	Fuel cost	70444.16	444.16	0.634518%
Ing-quote-1@0.1	CO2	310.533139	-10.066861	-3.140007%
Ing-quote-1@0.1	CH4	0.309856	0.304856	6097.115828%
Ing-quote-1@0.1	N2O	0.017011	-0.000989	-5.494805%
Ing-quote-1@0.1	MRV raw CO2e	323.716999	-1.793001	-0.550828%
Ing-quote-1@0.1	EU ETS CO2e before scope	323.716999	-1.793001	-0.550828%
Ing-quote-1@0.1	EUAs required	161.858500	-0.896500	-0.550828%
Ing-quote-1@0.1	EUA cost	12948.68	-71.72	-0.550828%
Ing-quote-1@0.1	Model cost	83392.84	372.44	0.448616%
Ing-quote-1@0.1	FuelEU scoped energy	2135.000	0.000	0.000000%
Ing-quote-1@0.1	FuelEU RWD denominator	4270.000	0.000	0.000000%
Ing-quote-1@0.1	WtT	14.8645	0.4645	3.225626%
Ing-quote-1@0.1	TtW	75.7257	-0.6417	-0.840325%
Ing-quote-1@0.1	GHGI	90.5902	-0.1772	-0.195273%
Ing-quote-1@0.1	Target	89.3368	0.0000	0.000000%
Ing-quote-1@0.1	Compliance balance (g)	-2676014.615828	378417.384172	12.389125%
Ing-quote-1@0.1	Compliance balance	-2.676015	0.378417	12.389125%
Ing-quote-1@0.1	Indicative penalty equivalent	1729.16	-240.67	-12.217709%

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
Ing-quote-1@0.1	Compliance improvement	0.378417	0.378417	-
Ing-quote-1@0.1	Reference-adjusted cost	-	-	-
Ing-quote-1@0.5	Baseline fuel mass	46.514	-53.486	-53.485839%
Ing-quote-1@0.5	Candidate fuel mass	46.514	46.514	-
Ing-quote-1@0.5	Physical energy	4270.000	0.000	0.000000%
Ing-quote-1@0.5	Fuel cost	72096.95	2096.95	2.995643%
Ing-quote-1@0.5	CO2	273.073012	-47.526988	-14.824388%
Ing-quote-1@0.5	CH4	1.444265	1.439265	28785.294118%
Ing-quote-1@0.5	N2O	0.013330	-0.004670	-25.941703%
Ing-quote-1@0.5	MRV raw CO2e	317.045005	-8.464995	-2.600533%
Ing-quote-1@0.5	EU ETS CO2e before scope	317.045005	-8.464995	-2.600533%
Ing-quote-1@0.5	EUAs required	158.522502	-4.232498	-2.600533%
Ing-quote-1@0.5	EUA cost	12681.80	-338.60	-2.600533%
Ing-quote-1@0.5	Model cost	84778.75	1758.35	2.117974%
Ing-quote-1@0.5	FuelEU scoped energy	2135.000	0.000	0.000000%
Ing-quote-1@0.5	FuelEU RWD denominator	4270.000	-0.000	-0.000000%
Ing-quote-1@0.5	WtT	16.5929	2.1929	15.228607%
Ing-quote-1@0.5	TtW	73.3377	-3.0297	-3.967286%
Ing-quote-1@0.5	GHGI	89.9307	-0.8368	-0.921912%
Ing-quote-1@0.5	Target	89.3368	0.0000	0.000000%
Ing-quote-1@0.5	Compliance balance (g)	-1267873.238562	1786558.761438	58.490703%
Ing-quote-1@0.5	Compliance balance	-1.267873	1.786559	58.490703%
Ing-quote-1@0.5	Indicative penalty equivalent	825.27	-1144.56	-58.104464%
Ing-quote-1@0.5	Compliance improvement	1.786559	1.786559	-
Ing-quote-1@0.5	Reference-adjusted cost	-	-	-
rfnbo-no-proof@0.1	Baseline fuel mass	90.000	-10.000	-10.000000%
rfnbo-no-proof@0.1	Candidate fuel mass	10.000	10.000	-
rfnbo-no-proof@0.1	Physical energy	4270.000	0.000	0.000000%
rfnbo-no-proof@0.1	Fuel cost	72000.00	2000.00	2.857143%
rfnbo-no-proof@0.1	CO2	320.600000	0.000000	0.000000%
rfnbo-no-proof@0.1	CH4	0.005000	0.000000	0.000000%
rfnbo-no-proof@0.1	N2O	0.018000	0.000000	0.000000%
rfnbo-no-proof@0.1	MRV raw CO2e	325.510000	0.000000	0.000000%
rfnbo-no-proof@0.1	EU ETS CO2e before scope	325.510000	0.000000	0.000000%
rfnbo-no-proof@0.1	EUAs required	162.755000	0.000000	0.000000%
rfnbo-no-proof@0.1	EUA cost	13020.40	0.00	0.000000%
rfnbo-no-proof@0.1	Model cost	85020.40	2000.00	2.409046%
rfnbo-no-proof@0.1	FuelEU scoped energy	2135.000	0.000	0.000000%
rfnbo-no-proof@0.1	FuelEU RWD denominator	4270.000	0.000	0.000000%
rfnbo-no-proof@0.1	WtT	14.4000	0.0000	0.000000%
rfnbo-no-proof@0.1	TtW	76.3674	0.0000	0.000000%
rfnbo-no-proof@0.1	GHGI	90.7674	0.0000	0.000000%
rfnbo-no-proof@0.1	Target	89.3368	0.0000	0.000000%
rfnbo-no-proof@0.1	Compliance balance (g)	-3054432.000000	0.000000	0.000000%

Scenario	Metric	Absolute	Delta	Relative-to-B0 (%)
rfnbo-no-proof@0.1	Compliance balance	-3.054432	0.000000	0.000000%
rfnbo-no-proof@0.1	Indicative penalty equivalent	1969.83	0.00	0.000000%
rfnbo-no-proof@0.1	Compliance improvement	0.000000	0.000000	-
rfnbo-no-proof@0.1	Reference-adjusted cost	-	-	-
rfnbo-no-proof@0.2	Baseline fuel mass	80.000	-20.000	-20.000000%
rfnbo-no-proof@0.2	Candidate fuel mass	20.000	20.000	-
rfnbo-no-proof@0.2	Physical energy	4270.000	0.000	0.000000%
rfnbo-no-proof@0.2	Fuel cost	74000.00	4000.00	5.714286%
rfnbo-no-proof@0.2	CO2	320.600000	0.000000	0.000000%
rfnbo-no-proof@0.2	CH4	0.005000	0.000000	0.000000%
rfnbo-no-proof@0.2	N2O	0.018000	0.000000	0.000000%
rfnbo-no-proof@0.2	MRV raw CO2e	325.510000	0.000000	0.000000%
rfnbo-no-proof@0.2	EU ETS CO2e before scope	325.510000	0.000000	0.000000%
rfnbo-no-proof@0.2	EUAs required	162.755000	0.000000	0.000000%
rfnbo-no-proof@0.2	EUA cost	13020.40	0.00	0.000000%
rfnbo-no-proof@0.2	Model cost	87020.40	4000.00	4.818093%
rfnbo-no-proof@0.2	FuelEU scoped energy	2135.000	0.000	0.000000%
rfnbo-no-proof@0.2	FuelEU RWD denominator	4270.000	0.000	0.000000%
rfnbo-no-proof@0.2	WtT	14.4000	0.0000	0.000000%
rfnbo-no-proof@0.2	TtW	76.3674	0.0000	0.000000%
rfnbo-no-proof@0.2	GHGI	90.7674	0.0000	0.000000%
rfnbo-no-proof@0.2	Target	89.3368	0.0000	0.000000%
rfnbo-no-proof@0.2	Compliance balance (g)	-3054432.000000	0.000000	0.000000%
rfnbo-no-proof@0.2	Compliance balance	-3.054432	0.000000	0.000000%
rfnbo-no-proof@0.2	Indicative penalty equivalent	1969.83	0.00	0.000000%
rfnbo-no-proof@0.2	Compliance improvement	0.000000	0.000000	-
rfnbo-no-proof@0.2	Reference-adjusted cost	-	-	-

Concrete switch points

Candidate	From scenario	To scenario	Value star (case currency)
uco-quote-1	B0	uco-quote-1@0.3	287.73
Ing-quote-1	B0	Ing-quote-1@0.5	984.21

FuelEU metrics and changes relative to B0

Scenario	WtT (g/MJ)	TtW (g/MJ)	GHGI (g/MJ)	Target (g/MJ)	Balance (tCO2e)	Indicative penalty equivalent (EUR)	Absolute / relative-to-B0
B0	14.4000	76.3674	90.7674	89.3368	-3.054432	1969.83	GHGI abs 0.0000; rel 0.000000%
uco-quote-1@0.2	0.8509	76.6720	77.5230	89.3368	25.222560	-	GHGI abs -13.2445; rel -14.591677%
uco-quote-1@0.3	-6.2062	76.8307	70.6244	89.3368	39.950878	-	GHGI abs -20.1430; rel -22.191879%
uco-quote-1@0.022 1306766829825081 0915896977272792 4816074006340564	12.9365	76.4003	89.3368	89.3368	0.000000	-	GHGI abs -1.4306; rel -1.576168%
Ing-quote-1@0.1	14.8645	75.7257	90.5902	89.3368	-2.676015	1729.16	GHGI abs -0.1772; rel -0.195273%

Scenario	WtT (g/MJ)	TtW (g/MJ)	GHGI (g/MJ)	Target (g/MJ)	Balance (tCO2e)	Indicative penalty equivalent (EUR)	Absolute / relative-to-B0
lng-quote-1@0.5	16.5929	73.3377	89.9307	89.3368	-1.267873	825.27	GHGI abs -0.8368; rel -0.921912%
rfnbo-no-proof@0.1	14.4000	76.3674	90.7674	89.3368	-3.054432	1969.83	GHGI abs 0.0000; rel 0.000000%
rfnbo-no-proof@0.2	14.4000	76.3674	90.7674	89.3368	-3.054432	1969.83	GHGI abs 0.0000; rel 0.000000%

Constraints and thresholds

Candidate	Max blend	Supply (t)	Budget	xBudget	xSupply	xCap	Target status	Target min	Target min cost	Max improvement	Cost min
uco-quote-1	30.0000	-	-	100.0000	100.0000	30.0000	TARGET_REACHABLE	2.2131	2.2131	30.0000	0.0000
lng-quote-1	50.0000	-	-	100.0000	100.0000	50.0000	TARGET_UNREACHABLE_UNDER_CONSTRAINTS	-	-	50.0000	0.0000
rfnbo-no-proof	20.0000	-	-	100.0000	100.0000	20.0000	TARGET_NO_SOLUTION	-	-	-	0.0000

Factor evidence and resolution

Requested path	Resolved path	Fallback reason	Qualification	Factor status	Evidence	Values
MDO	MDO	DIRECT_RESOLUTION	NOT_DEMONSTRATED	FIXED	FACTOR-CATALOG:MDO	lcv_mj_per_g=0.0427; wt_t_g_per_mj=14.4; cf_co2_g_per_g=3.206; cf_ch4_g_per_g=0.00005; cf_n2o_g_per_g=0.00018; rwd=1
UCO_FAME	UCO_FAME	DIRECT_RESOLUTION	NOT_DEMONSTRATED	ESTIMATED	FACTOR-CATALOG:UCO_FAME	lcv_mj_per_g=0.037; wt_t_g_per_mj=-61.694594595; cf_co2_g_per_g=2.834; cf_ch4_g_per_g=0.00005; cf_n2o_g_per_g=0.00018; rwd=1
LNG_OTTO_MEDIUM_SPEED	LNG_OTTO_MEDIUM_SPEED	DIRECT_RESOLUTION	NOT_DEMONSTRATED	FIXED	FACTOR-CATALOG:LNG_OTTO_MEDIUM_SPEED	lcv_mj_per_g=0.0491; wt_t_g_per_mj=18.5; cf_co2_g_per_g=2.75; cf_ch4_g_per_g=0; cf_n2o_g_per_g=0.00011; rwd=1; csliip_percent=3.1
E_DIESEL	MDO	RFNBO_QUALIFICATION_NOT_DEMONSTRATED	NOT_DEMONSTRATED	FIXED	FACTOR-CATALOG:MDO	lcv_mj_per_g=0.0427; wt_t_g_per_mj=14.4; cf_co2_g_per_g=3.206; cf_ch4_g_per_g=0.00005; cf_n2o_g_per_g=0.00018; rwd=1

Port evidence

Role	Port	UN/LOCODE	EU ETS identity	FuelEU identity	Rule source ID	Source version
Departure	Shanghai Pt	CNSHG	THIRD_COUNTRY	THIRD_COUNTRY	PORT-02;ETS-01;FEU-01	UN/LOCODE 2025-1
Arrival	Rotterdam	NLRTM	IN_SCOPE	IN_SCOPE	PORT-02;ETS-01;FEU-01	UN/LOCODE 2025-1

Issues

Scope	Candidate	Scenario	Component	Code	Field	Blocking	Message
-	-	-	-	-	-	-	No issues recorded

Methodology and versions

Method	Value
Calculation specification version	2026-08-07
Fuel factor version	2026-08-31-audit
Port rule version	2026-07-23
Calculation status vocabulary	BLOCKED, CALCULABLE, COMPARABLE
Execution status	EXECUTION_CONDITIONS_PENDING
FuelEU limitation	Voyage-level proportional estimate; indicative EUR equivalent is not an annual legal penalty or annual-limit result.